Walid Birouk

AI & Data Engineer

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SUMMARY

AI & Data Engineer with hands-on experience in generative AI (RAG, LLM agents). I have built AI assistants integrated into business tools, ML systems for railway safety, and the data model and automations of a production e-commerce store.

EXPERIENCE

Ghazeline | Data Engineer & Maintainer — Jan 2026 – Present

- Designed the **PostgreSQL / Supabase** data model for an online boutique of a women's fashion brand with 30+ products.
- Deployed an **n8n AI agent** that answers the store's **50–100 daily Instagram customer messages**, so the team only steps in for complex requests.
- Built backend data-collection features, automated order summaries and cash-on-delivery confirmation, and maintain the store in production from business feedback.

Infrabel (RideOnTrack project) | ML Engineer — Jan 2025 – Aug 2025

- Developed an anomaly detection system for Belgian railway tracks with machine learning and computer vision, using a U-Net segmentation model and an autoencoder to isolate critical areas, enabling targeted inspection.
- Integrated YOLOv8 for real-time train signal recognition (91% accuracy).

Algorhythm — Feb 2024 – Dec 2024

Generative AI Engineer — Sep 2024 – Dec 2024

- Built a **RAG agent** on **Vertex AI** over internal **BigQuery** data and integrated it into **HubSpot** for about 50 employees, so they get answers from company data without leaving their CRM.
- Developed Vertex AI workflows that automate recurring internal writing tasks in a consistent, specialised tone.

Data Engineering & Data Quality Intern — Feb 2024 – Jul 2024

- Built a data observability proof of concept: automated dbt tests and manual data-quality checks, with results piped to OpenMetadata, giving visibility into data quality.

SKILLS

AI / ML: Generative AI, RAG, LLM agents, TensorFlow, YOLOv8, U-Net, autoencoders
Languages & data: Python, SQL, PostgreSQL, Supabase, BigQuery, dbt, Azure Databricks
Data engineering: PySpark, Spark SQL, Airflow
Cloud & MLOps: Google Cloud (Vertex AI), AWS (SageMaker), Azure, Docker, Kubernetes (kind), FastAPI, Dataiku, Git
Other: HubSpot integration, requirements analysis, data modelling, production support

EDUCATION

Thomas More, Mechelen, Belgium | Bachelor's in Data & Artificial Intelligence — 2021 – 2024

University of Jijel | Mathematics & Computer Science (Licence) — 2019 – 2021

CERTIFICATIONS & LANGUAGES

- AWS Academy Graduate – Data Engineering
- Microsoft: Azure Databricks for Data Engineering
- dbt Labs: dbt Fundamentals
- Dataiku: Generative AI Practitioner, ML Practitioner, MLOps Practitioner, Core Designer, Advanced Designer, Developer
- IBM: Machine Learning with Python

See credentials on LinkedIn

Languages: English: Fluent, French: Professional working proficiency, Dutch: Basic (A2), Amazigh: Native